PROJECT GUIDE

Orbi

The connection popup Windows never had

Earbuds, Wi-Fi, Ethernet and USB drives. A small card slides up the moment something connects, then goes away. Orbi starts with Windows and tells you what is already connected.



A real Orbi popup captured on the author's PC: Wi-Fi, 300 Mbps, excellent signal.

MADE BY
Gopinath

VERSION
0.1.0

PLATFORM
Windows 11

BUILT WITH
Tauri 2 + Rust

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1.4 MB
installer

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SECTION 01

Why Orbi exists

A MacBook shows a card the moment AirPods connect. Windows shows a short sound and a small toast that is easy to miss. Orbi gives Windows one clear popup for any kind of connection, at almost no cost in CPU or memory.

One glance

See which earbuds connected and the battery of each bud and the case, without opening Settings.

Any connection

Earbuds, Wi-Fi, Ethernet and USB drives share one popup, with a layout for each.

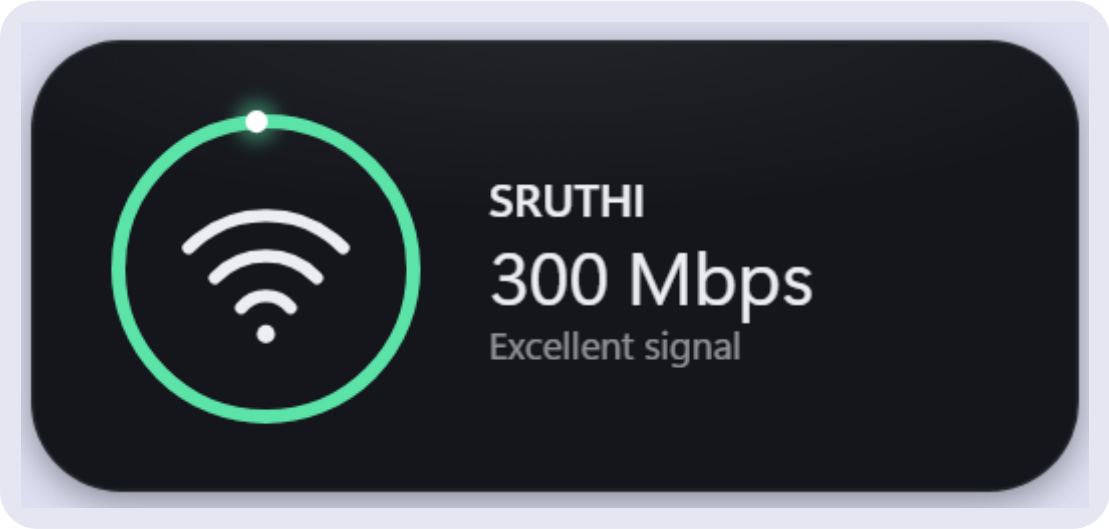
Out of the way

The popup shows for about 5 seconds and is then destroyed, so nothing sits on screen.

Local and light

No account, no cloud, no polling. Windows tells Orbi when something changes.

The real thing



A real popup from the installed app: Orbi reading the PC's own Wi-Fi. Size on screen is 380 by 270 at the bottom centre.

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SECTION 02

Every feature

Everything below ships in this repository today.

Devices

- Bluetooth earbuds: battery from Windows, with the full pair and case drawn when only one value is reported.
- AirPods: left, right and case battery read from Apple's Bluetooth advertisements (unit tested, not yet confirmed on real AirPods).
- Wi-Fi: network name, signal strength and link speed.
- Ethernet: connected adapter and link speed.
- USB drives: drive letter, how full it is and free space (removable drives only).

Behaviour

- Popup on connect, and again when a battery newly drops to 20% or below.
- A device that reconnects within 2 seconds does not pop up twice (connection flapping).
- Hover over the popup to keep it open; use the close button to dismiss it.
- The tray tooltip lists every connected device and its battery.
- Per-type switches in the tray menu, remembered between runs.

On launch

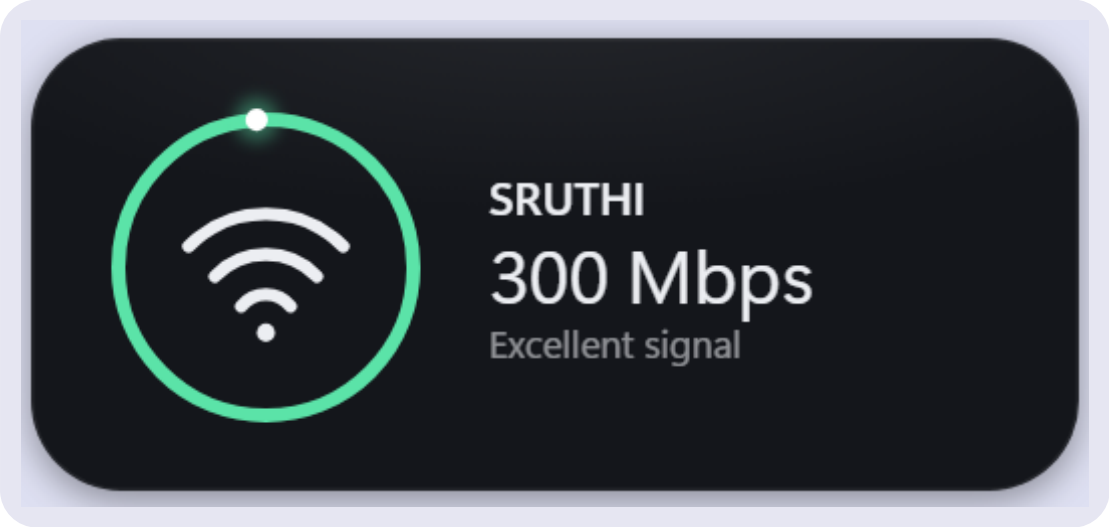
- Starts with Windows by default on first run. You can turn it off in the tray menu.
- When Orbi opens, it checks what is already connected and shows one popup for the most useful device: earbuds, then Wi-Fi, Ethernet, USB.
- After that, every new connection shows a popup automatically.

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SECTION 03

How it looks: every layout

The layout is chosen from the device type. All share the Orbit design: a ring that fills with the value and a glowing dot.



Wi-Fi layout: network name, link speed, signal quality, ring filled by signal strength.

Type	Shows	Read from
Earbuds	Name, left and right bud, case, battery rings, charging bolt	Windows device battery property
AirPods	Same, with real left, right and case values	Apple Bluetooth advertisements
Wi-Fi	Network name, speed, signal quality	WLAN API and interface table
Ethernet	Adapter name, link speed	Interface table
USB drive	Drive letter, used and free space	Volume notifications and drive info

Colour tells the story: a ring turns amber then red as battery falls, and a device at 20% or below adds a Low battery tag. Artwork is SVG (src/art.ts) with a lookup by device name, so a known product can get its own look.

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SECTION 04

Tech stack

Small on purpose: a native shell with a web page used only for the popup.

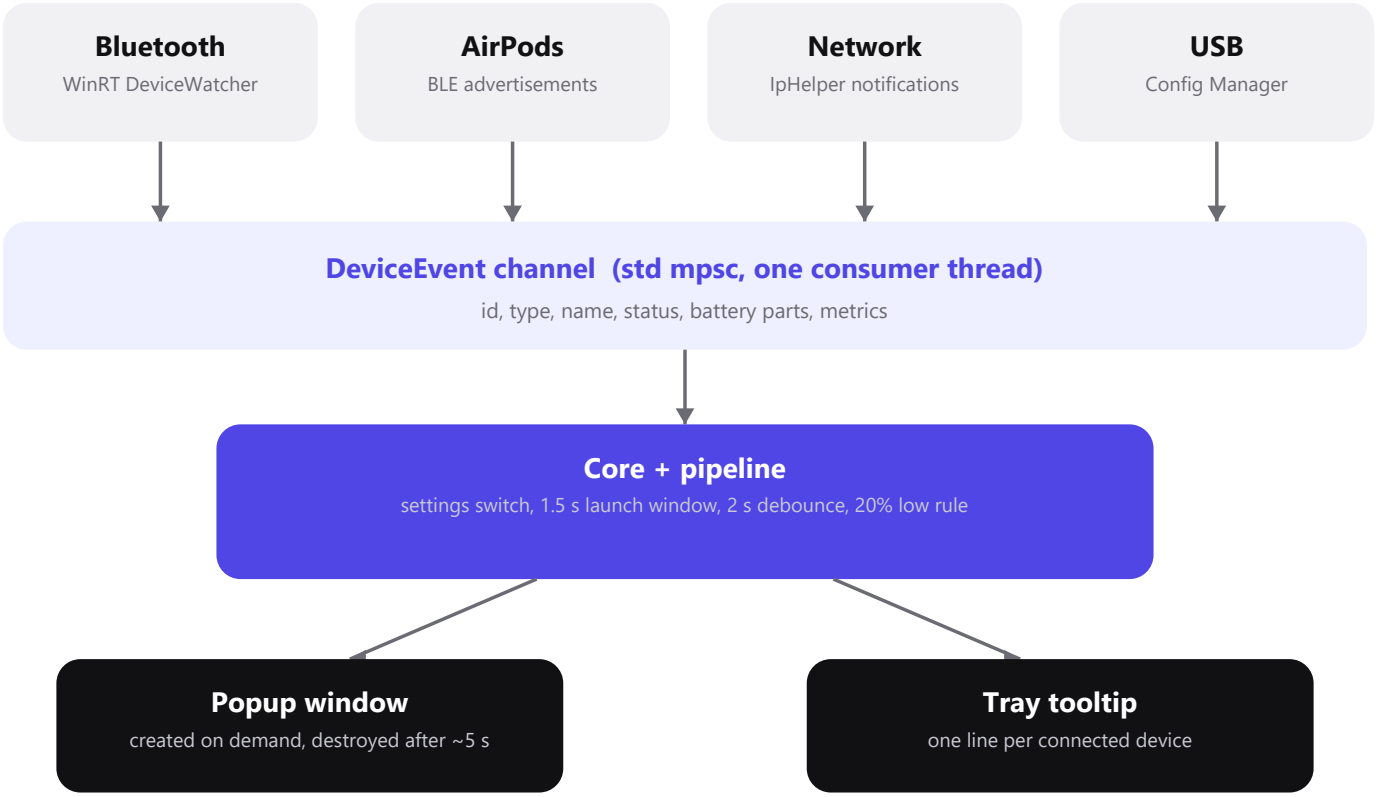
Piece	Choice	Why
Shell	Tauri 2 (Rust)	Tiny installer, low idle memory, tray support
Popup UI	TypeScript, Vite, SVG	Fast to draw, one small page
Windows APIs	windows crate 0.62	Direct access to Bluetooth, network and volume notifications
Settings	JSON file	Simple and easy to inspect
Autostart	tauri-plugin-autostart	Registers Orbi to start with Windows
Installer	NSIS	About 1.4 MB

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SECTION 05

Architecture

Watchers push events into one core. The core decides what is worth showing.



Each watcher subscribes to a Windows notification and sleeps until it fires. Then it reads the device again, compares with the last read and sends only what changed. One consumer thread owns the core, so there are no locks on the device state.

Why this shape

- No polling: idle CPU is near zero because every thread is blocked waiting on Windows.
- One event type: the core and the popup never care which watcher spoke.
- The core is plain Rust with no Windows calls, so it is fully unit tested.
- The popup window exists only while visible, so idle memory is just the tray process.

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SECTION 06

How each part works

One module per job in src-tauri/src.

File	Job
lib.rs	Builds the tray menu, applies first-run autostart, starts the pipeline.
events.rs	The normalized DeviceEvent: id, type, name, status, battery parts, metrics.
core.rs	Live-device state, 2 s debounce, 20% low-battery rule, launch popup choice.
pipeline.rs	Connects watchers to core to popup. Holds launch events for 1.5 s and shows one popup.
popup.rs	Creates the popup window, closes it after about 5 s, honours hover and close.
settings.rs	Reads and writes settings.json, per-type switches, popup time.
watchers/bluetooth.rs	Connected Bluetooth devices and battery; ignores switched-off paired devices.
watchers/airpods.rs	Parses Apple advertisements into left, right and case levels.
watchers/network.rs	Wi-Fi and Ethernet via network interface change notifications.
watchers/usb.rs	Removable drives via volume interface notifications.
mock.rs	Test fixtures for the unit tests. Debug builds only; release builds never show fake devices.

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SECTION 07

Windows endpoints

Orbi has no server and calls no web API. Every value comes from a Windows interface on this PC.

Watcher	Wakes on	Reads
Bluetooth	WinRT DeviceWatcher over BluetoothDevice.GetDeviceSelectorFromConnectionStatus(Connected)	Battery: DEVPKEY 104EA319-6EE2-4701-BD47-8DDBF425BBE5, pid 2. Connected flag: 83DA6326-97A6-4088-9453-A1923F573B29. Name: DEVPKEY_Device_FriendlyName
AirPods	BluetoothLEAdvertisementWatcher, Apple company id 0x004C	Proximity-pairing payload: left, right, case, charging bits. Ignored below -60 dBm
Wi-Fi	iphlpapi NotifyIpInterfaceChange (AF_UNSPEC)	wlanapi WlanOpenHandle, WlanEnumInterfaces, WlanQueryInterface: SSID, signal quality, receive rate
Ethernet	iphlpapi NotifyIpInterfaceChange	iphlpapi GetIfTable2: adapter alias, operational state, link speed
USB	cfgmgr32 CM_Register_Notification on GUID_DEVINTERFACE_VOLUME {53F5630D-B6BF-11D0-94F2-00A0C91EFB8B}	kernel32 GetLogicalDrives, GetDriveTypeW (DRIVE_REMOVABLE), GetVolumeInformationW, GetDiskFreeSpaceExW

Quiet periods

After a notification each watcher waits for the burst to settle before it reads: Bluetooth 1.5 s, USB 0.9 s, network 0.6 s. Bluetooth also re-reads every 60 s while a device is connected, because Windows does not announce battery changes.

Local files

Settings: %APPDATA%\com.orbi.app\settings.json. Autostart: the current user's Run key, written by tauri-plugin-autostart. Nothing else is read or written.

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SECTION 08

Repository map

github.com/goprocker/Orbi, with what each file is for.

orbi/	
src/	popup UI, runs inside WebView2
main.ts	layout per device type, ring and battery logic
art.ts	SVG artwork for buds, case, Wi-Fi, USB
styles.css	Orbit design: glass card, rings, colours
src-tauri/	native Rust app
src/lib.rs	tray menu, first-run autostart, wiring
src/pipeline.rs	watchers to core to popup, launch window
src/core.rs	state, debounce, low-battery rule, tests
src/events.rs	DeviceEvent, DeviceType, Part, Metrics
src/popup.rs	popup window lifecycle, WebView2 flags
src/settings.rs	settings.json load and save
src/watchers/	bluetooth, airpods, network, usb
tauri.conf.json	app id com.orbi.app, NSIS installer
icons/	Orbi icon set
index.html	popup page entry
app-icon.png	icon source
Orbi-Tech-Plan.pdf	plan and progress log
Orbi-Guide.pdf	this guide
README.md	short status and run steps

Rust to UI: the core sends a DeviceEvent to the popup window as a JSON event named device. The popup calls popup_ready, popup_hover and popup_close back into Rust.

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SECTION 09

Install and run

Windows 11 only.

Requirements to build

- Rust (stable) and the Visual Studio C++ Build Tools (Desktop development with C++).
- Node.js and npm.
- WebView2 (included with Windows 11).
- Windows Smart App Control must be off, because it blocks unsigned Rust build helpers.

Build from source

```
npm install
```

```
npm run tauri build
```

Use npm run tauri build, not a plain cargo build. A plain cargo build makes the popup load the dev server and show a blank error page. The installer is a per-user install, so it needs no admin rights.

Install

Run npm run tauri build. The installer is written to src-tauri/target/release/bundle/nsis/Orbi_0.1.0_x64-setup.exe. Run it and Orbi starts in the tray.

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SECTION 10

Using Orbi

There is no main window. Orbi lives in the tray.

First launch

- Orbi starts and adds itself to Windows startup (first run only).
- It checks what is already connected and shows one popup within about two seconds.
- From then on, connecting earbuds, joining a network, plugging in Ethernet or inserting a USB drive shows a popup.

Tray menu

- Check items for Earbuds and Bluetooth, Wi-Fi, Ethernet and USB drives.
- Start with Windows.
- Quit Orbi.

Launch timeline



The tooltip

Hover the tray icon to see every connected device with its battery, or 'Orbi: no devices'.

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SECTION 11

Settings

Stored in %APPDATA%\com.orbi.app\settings.json. Missing or broken values fall back to defaults.

Setting	Default	Meaning
bluetooth	true	Show popups for earbuds and Bluetooth devices
wifi	true	Show popups for Wi-Fi
ethernet	true	Show popups for Ethernet
usb	true	Show popups for USB drives
popup_seconds	5	How long the popup stays (clamped to 2 to 30)
autostart_initialized	false	Set after the first run so turning autostart off sticks

Disabled types still pass disconnect events, so Orbi's live state stays correct when you switch them back on.

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SECTION 12

Build, test, debug

Tests

cd src-tauri, then cargo test. 24 unit tests cover the core rules, settings, AirPods parsing, network and Bluetooth logic and the launch popup choice. They use fixture data in memory; the app itself never shows fake devices.

Check it on real devices

- Connect earbuds: the popup shows their battery.
- Join or switch a Wi-Fi network: the popup shows its name and speed.
- Plug in or remove a USB drive, or a network cable.
- Close and reopen Orbi: it shows one popup for what is already connected.

Logs

A debug build prints lines starting with [orbi] to the console, for example which watchers started and any popup or autostart error.

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SECTION 13

Privacy, troubleshooting, uninstall

Privacy

Orbi reads device names, battery levels, the network name and drive space on your PC only to draw the popup. It sends nothing anywhere and has no account.

Troubleshooting

Problem	Try
Popup does not appear	Check the tray switch for that device type is ticked.
Build is blocked	Turn Windows Smart App Control off.
Blank popup in a dev build	Use npm run tauri build instead of a plain cargo build.
Wrong earbuds shown	Orbi ignores paired devices that are switched off. Reconnect the earbuds.
AirPods left and right swapped	Known risk until verified on real AirPods.

Uninstall

Uninstall Orbi from Windows Settings, Apps. Settings live in %APPDATA%\com.orbi.app and can be deleted by hand.

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SECTION 14

Hard problems and limits

What was difficult and what is still open.

Solved

- Windows keeps the last battery value of switched-off paired earbuds. Orbi only trusts values from devices Windows reports as connected.
- WebView2 normally uses about 175 MB. Tuned launch flags bring the popup helper to about 82 MB in a debug build test.
- Mica or acrylic was rejected: it fills the rectangular window and ignores the card's rounded corners.

Known limits

- USB hard drives and SSDs are not detected, because Windows reports them as fixed disks.
- Bluetooth and AirPods code needs real hardware to verify fully.
- The release build with the tuned flags has not been re-measured.
- At launch only one popup is shown, even if several devices are connected.

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SECTION 15

Words you need

Word	Meaning
Tray	The icons by the clock where Orbi lives.
Watcher	A module that listens to Windows for one kind of change.
Event	A small record: which device, what happened, battery and metrics.
Debounce	Ignoring a repeat that comes too soon after the last one.
WebView2	The browser engine Windows provides to draw the popup page.
Tauri	A framework for desktop apps with a Rust core and a web UI.
Autostart	Starting a program automatically when Windows starts.

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SECTION 16

Cheat sheet

Want to	Do this
Build the app	npm install; npm run tauri build
Run the tests	cd src-tauri; cargo test
Find the installer	src-tauri/target/release/bundle/nsis/Orbi_0.1.0_x64-setup.exe
Stop it starting with Windows	Untick Start with Windows in the tray menu
Change how long it shows	Edit popup_seconds in settings.json

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